

Uber Particle Shader (English)

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Tags	Products
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Particle Feature (Base)

▼ MainTex



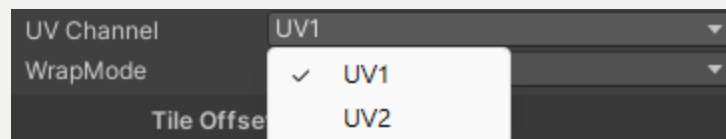
Function Info

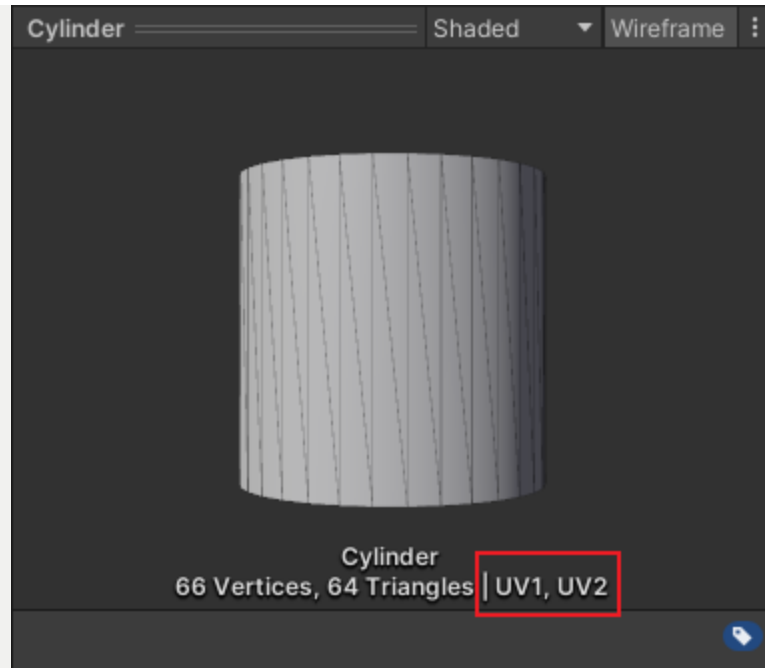
▼ MainTex Mode



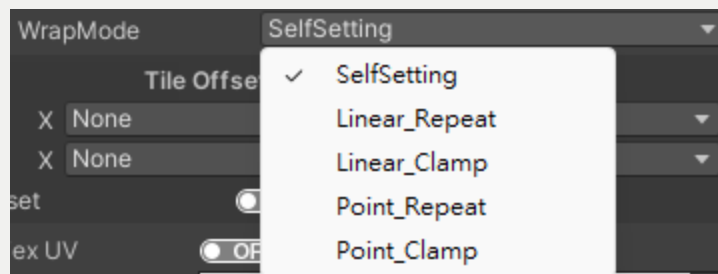
Function	Introduce	Example
Normal	Standard Usage Mode	None
MultiChannel	Multi-Color Palette, allows for texture variation through fading in and out of textures	Multi-Channel Tex
FlipBook	Animated Sequence Textures	FlipBook
PolarCoord	UV Polar Coordinates, enables UV expansion from the center outward in a circular pattern	Polar Coord
ScreenUV	Full-Screen UV, allows UVs to cover the entire screen for special effects	ScreenUV

▼ UV Channel - Samples textures based on the UVs stored in the current model, **with support for up to two UV sets.**

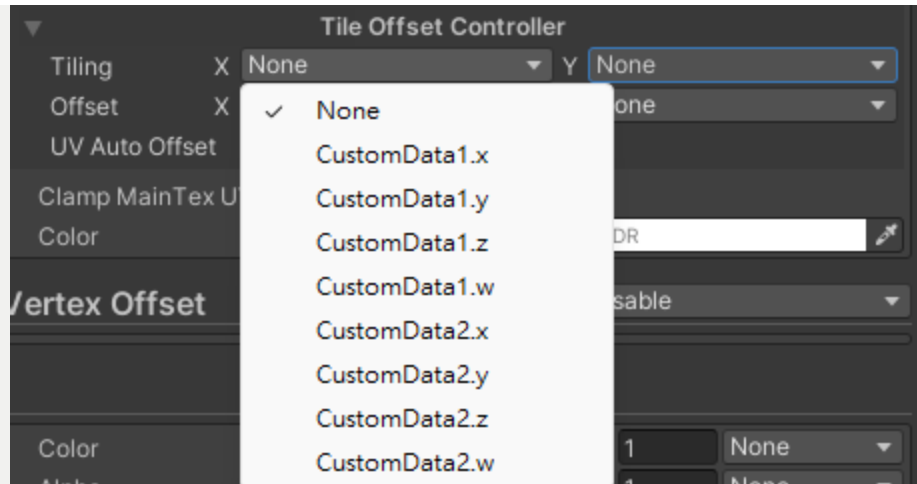




▼ WrapMode - Allows selection of the sampling method for textures **(not applicable with OpenGL API)**.



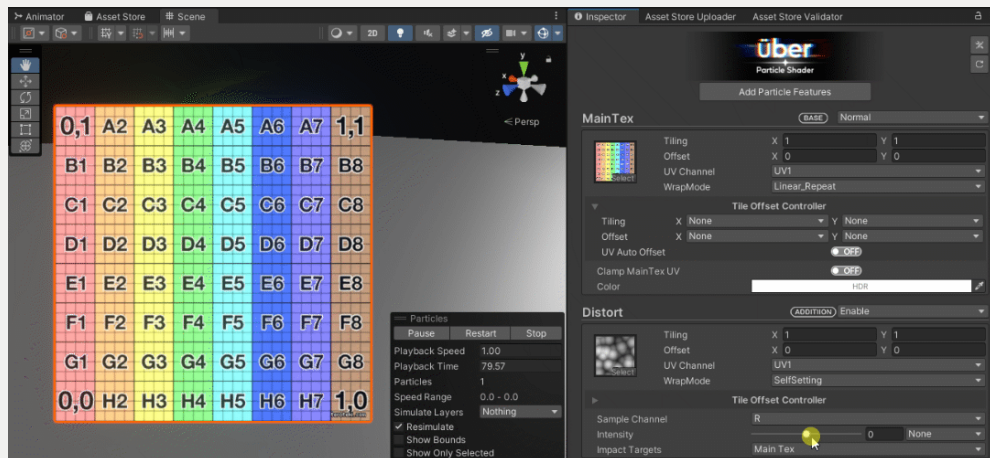
▼ Tile Offset Controller - Enables each parameter to be controlled by a specific Particle's Custom Data.



▼ Clamp MainTex UV - Clips the UVs if the X or Y value exceeds 1 or falls below 0.



When using the Distort function, it is easy for UVs to exceed the original boundaries and create excess pixels; this feature can be used to clip the excess.



Particle Feature (Additional)

▼ Distort



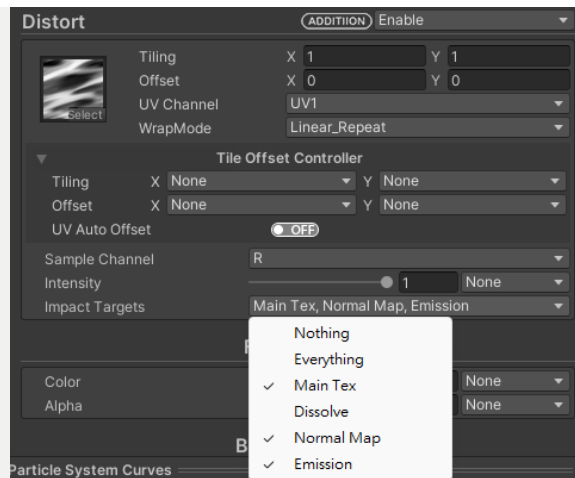
Function Info

▼ Distort Mode

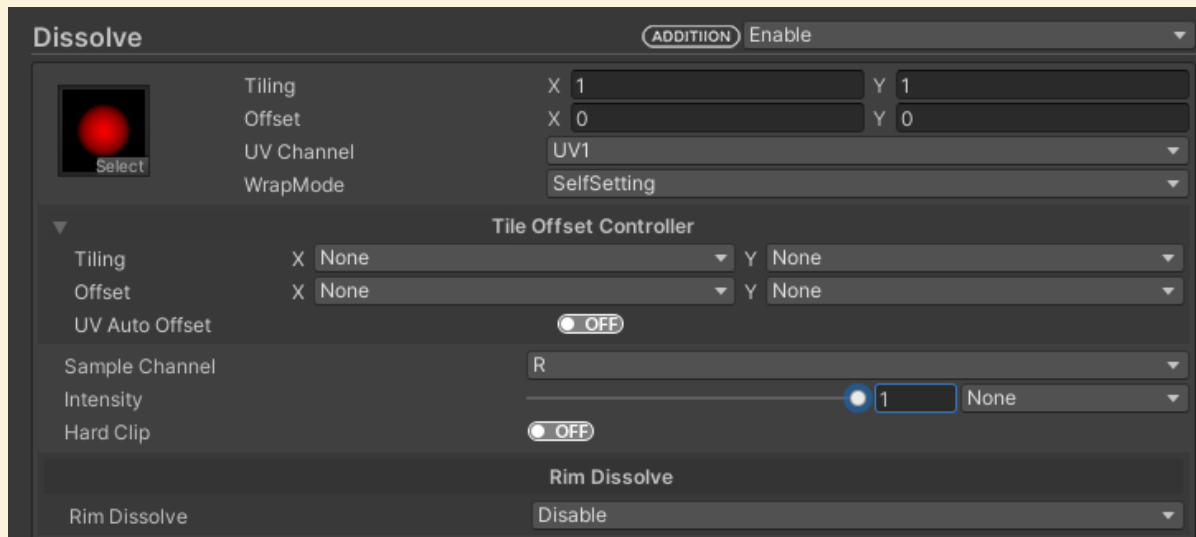


Function	Introduce	Example
Enable	Standard Usage Mode	None
PolarCoord	UV Polar Coordinates, allows for expansion of UVs from the center outward in a circular pattern	Polar Coord
FlowMap	Vector Flow Map, allows UVs to be influenced by the RG channels	Flow Map
ScreenUV	Full-Screen UV, expands UVs to cover the entire screen for special effects	ScreenUV

- Sample Channel - Selects which channel of the texture to sample for use
- ▼ Impact Targets - Select which effects to apply to (multiple selections possible)

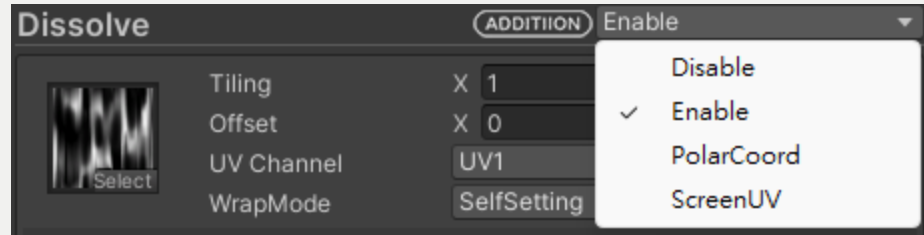


▼ Dissolve



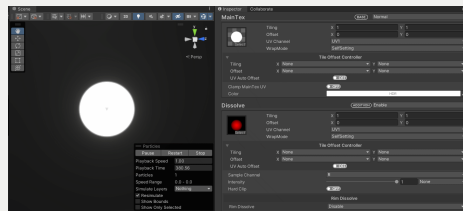
Function Info

▼ Dissolve Mode

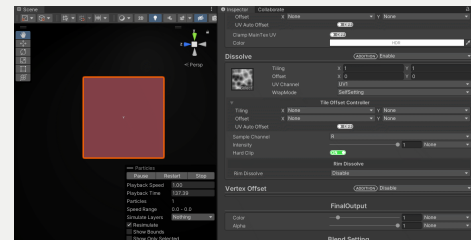


Function	Introduce	Example
Enable	Standard Usage Mode	None
PolarCoord	UV Polar Coordinates, allows for expansion of UVs from the center outward in a circular pattern	Polar Coord
ScreenUV	Full-Screen UV, expands UVs to cover the entire screen for special effects	ScreenUV

▼ Hard Clip



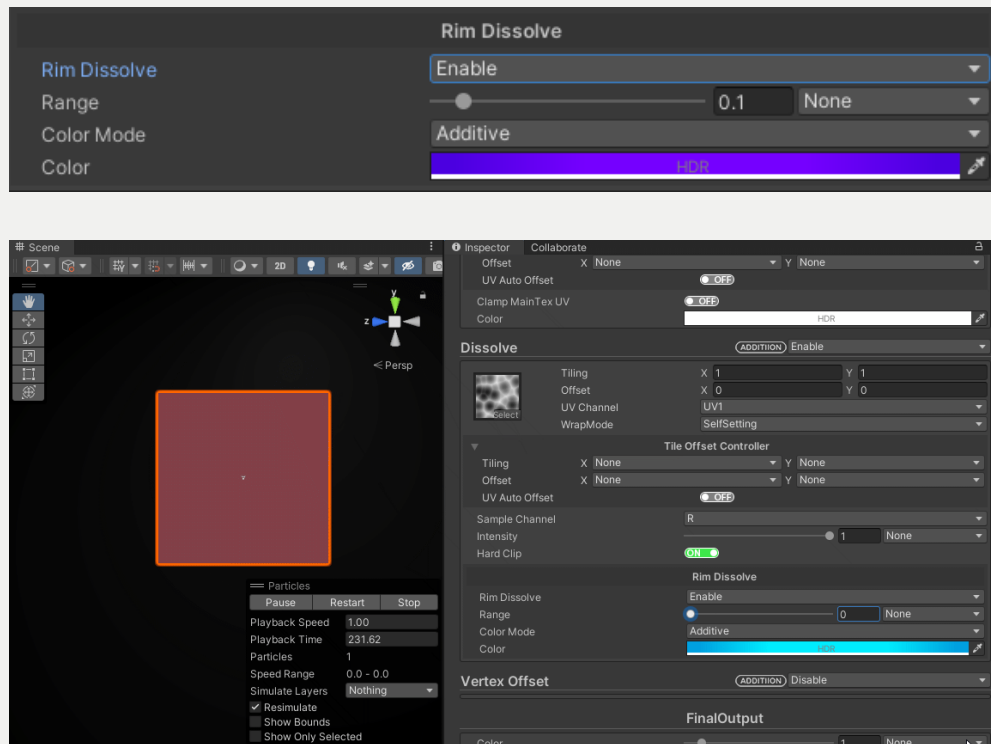
Example 1



Example 2

⚠ When the Hard Clip function is used, the Intensity that normally controls the strength of disappearance is **clamped between 0 and 1. This means the original dissipation range of $[-1, 1]$ is compressed to $[0, 1]$.**

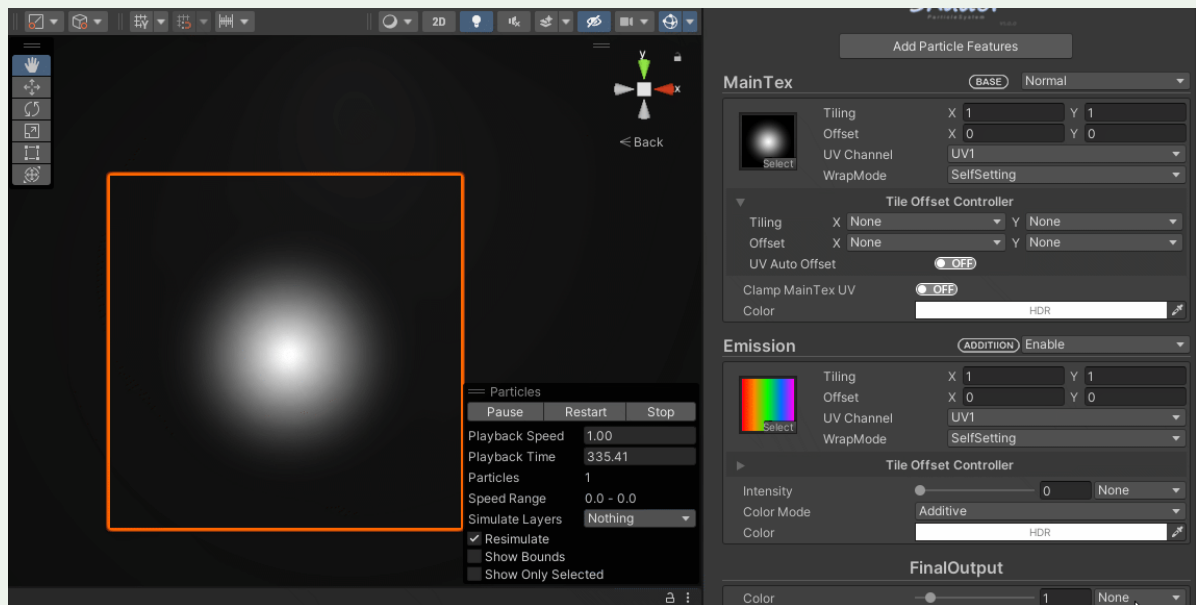
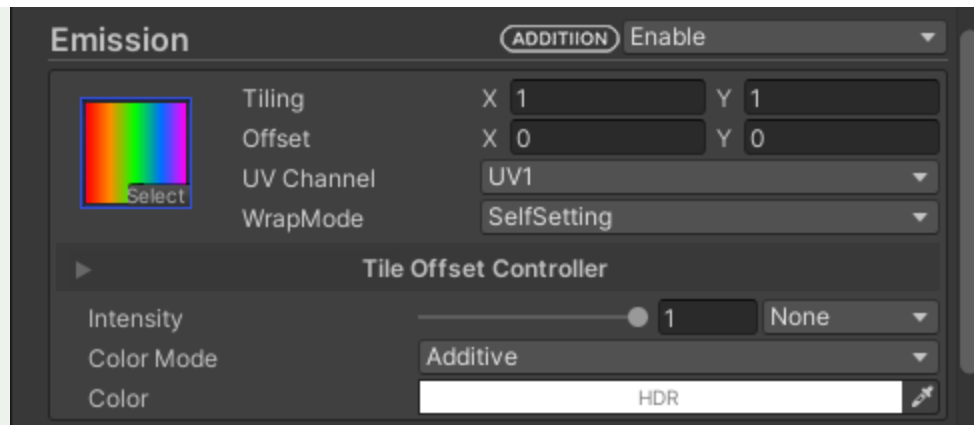
▼ Rim Dissolve



Example

- **Range** - Determines the range of the rim dissolution. The larger the value, the larger the range of the halo, and vice versa.
- **Color Mode** - Additive, Replace, Multiply.
- **Color** - The color of the rim.

▼ Emission

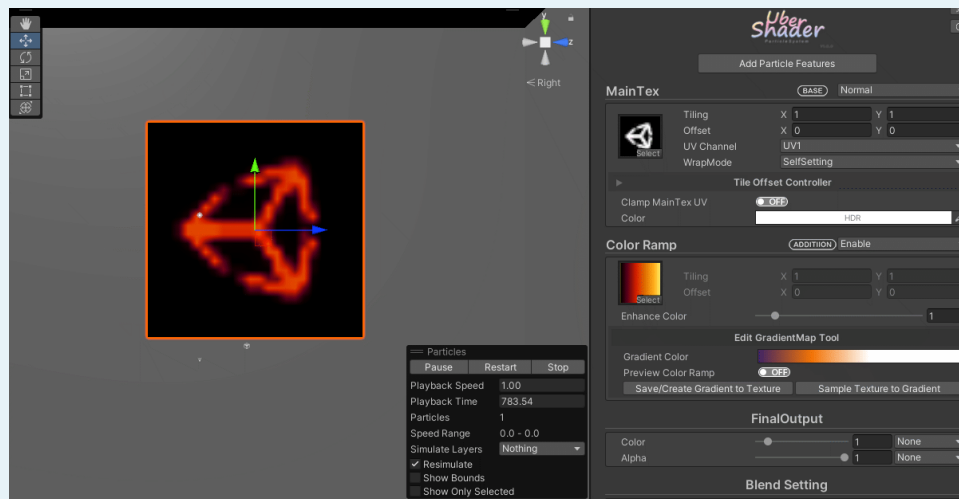
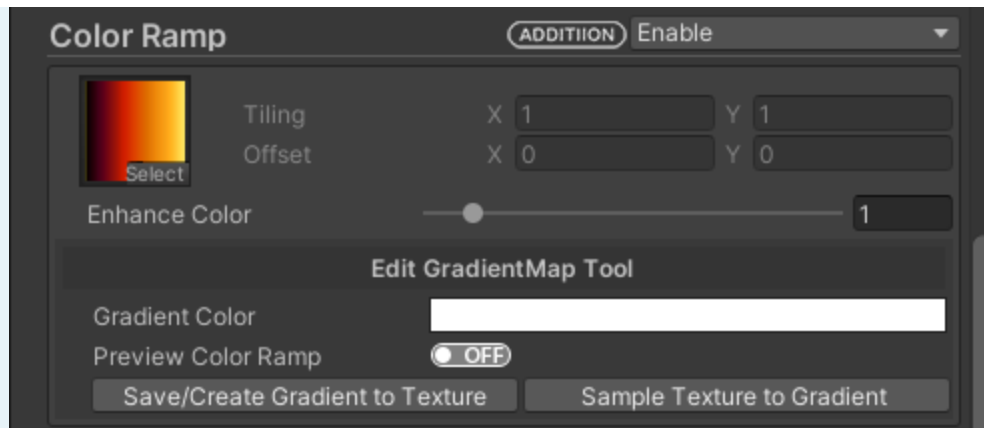


Example

Function Info

- **Intensity** - Controls the overall intensity of Emission.
- **Color Mode** - Additive (brightens), Multiply (overlays), Replace (replaces).
- **Color** - Controls the overall color of Emission.

▼ ColorRamp



Example



Function Info

- **Enhance Color** - Enhances the overall color intensity.



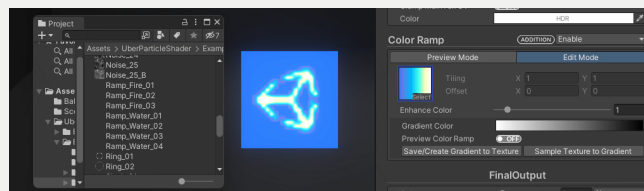
Edit GradientMap Tool - Tool for editing gradient maps.

- ▼ Preview Color Ramp - Live preview of the Gradient Color effects.

1

Automatic Texture Creation

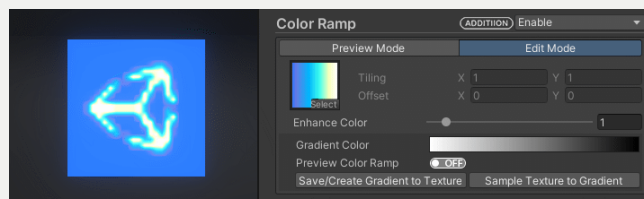
- When this feature is activated, the area originally designated for a texture is forcibly replaced with a temporarily created GradientMap. Upon deactivation, the original texture is restored automatically.



2

Real-Time Color Replacement

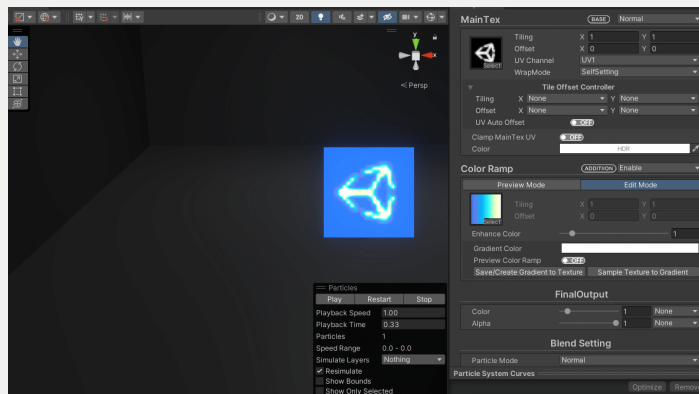
- When this feature is enabled, editing the Gradient Color directly allows for the alteration of corresponding colors on the texture.



▼ Save / Create Gradient to Texture - Save or create Gradient Color as a texture.



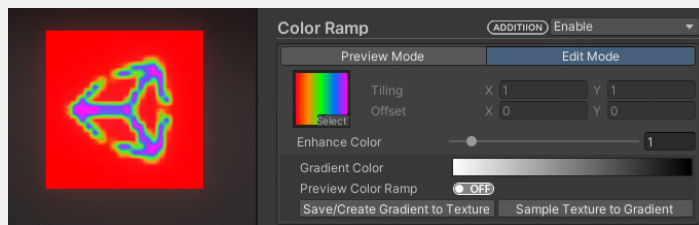
After finalizing the Gradient Color adjustments, pressing this button will initiate a save process (a save interface will appear). The finalized image will be **automatically replaced** in the project.



▼ Sample Texture to Gradient - Samples the gradient information from a GradientMap created within this tool.

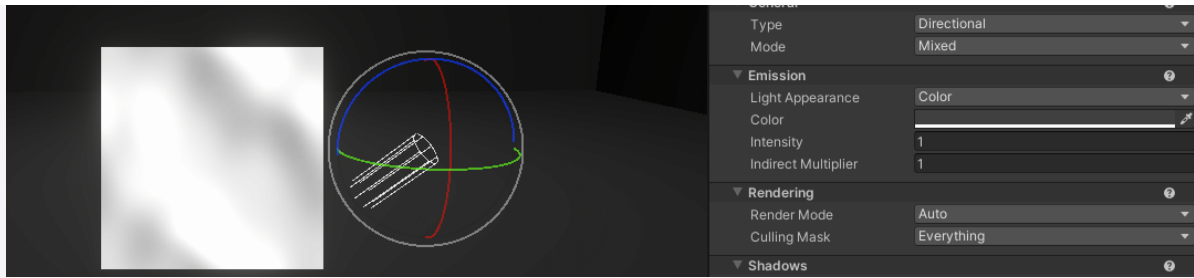


This feature **is limited to GradientMaps created by this tool** and allows direct sampling of the original gradient information from the texture. When exporting and importing into other projects using PackageManager, the data of sampling gradients is lost. It is recommended to package the entire Meta file for sampling capabilities.



▼ NormalMap



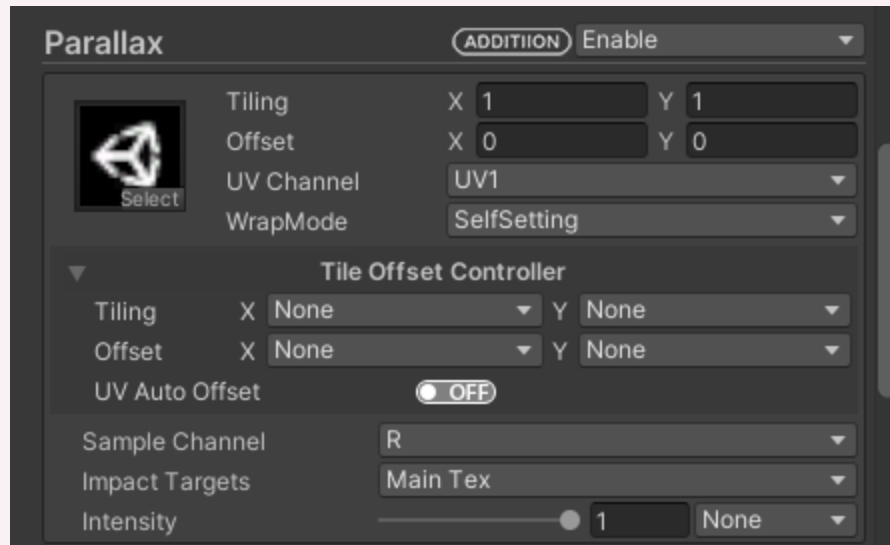


Example

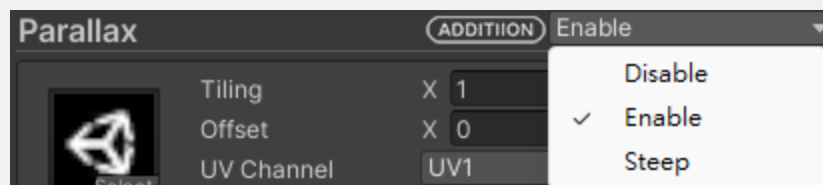
Function Info

- **Intensity** - Controls the overall intensity of Normal.
- **Bounce Light Range** - Controls the threshold for diffuse reflected light.

▼ Parallax



▼ Parallax Mode

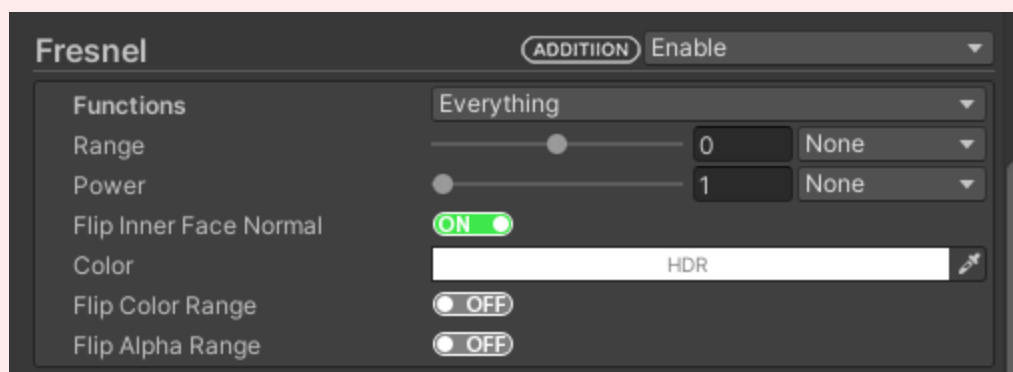




Function Info

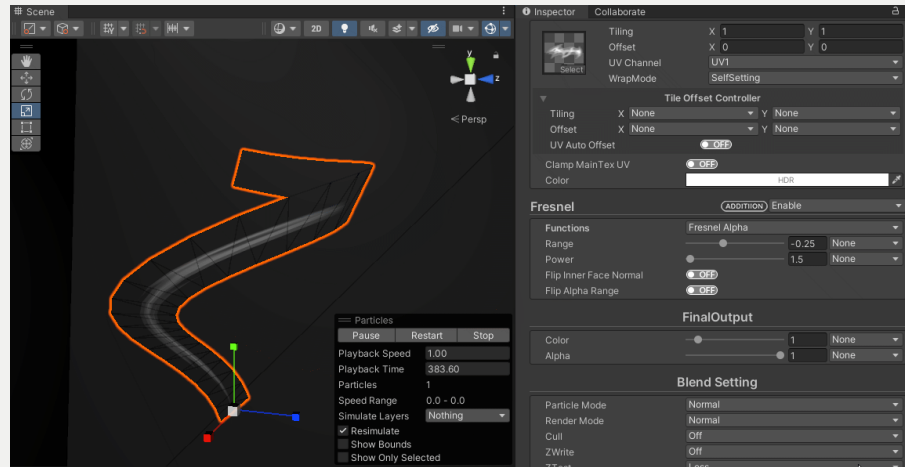
Function	Introduce	Example
Enable	Standard Usage Mode	Parallax (Enable)
Steep	Raystep-Sampling Mode	Parallax (Steep)

▼ Fresnel

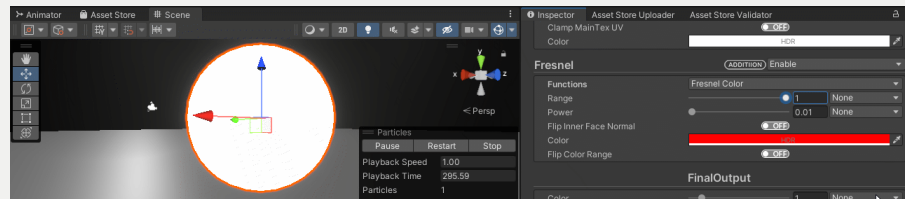


Function Info

- 1 **Flip Inner Face Normal** - Reverses the normals on the inside of the model. (Allows correct texture display on both sides in double-sided rendering.)

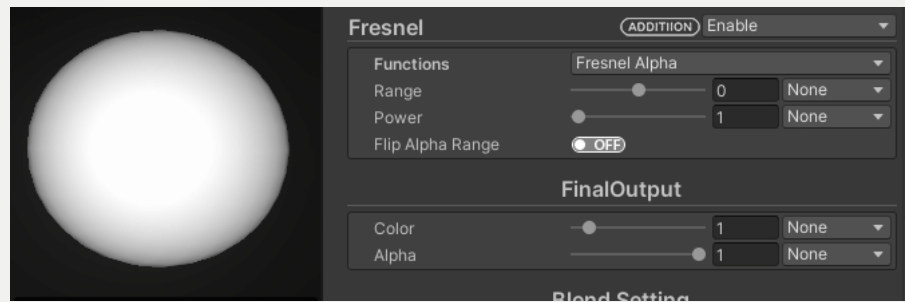


- 2 **Fresnel Color**



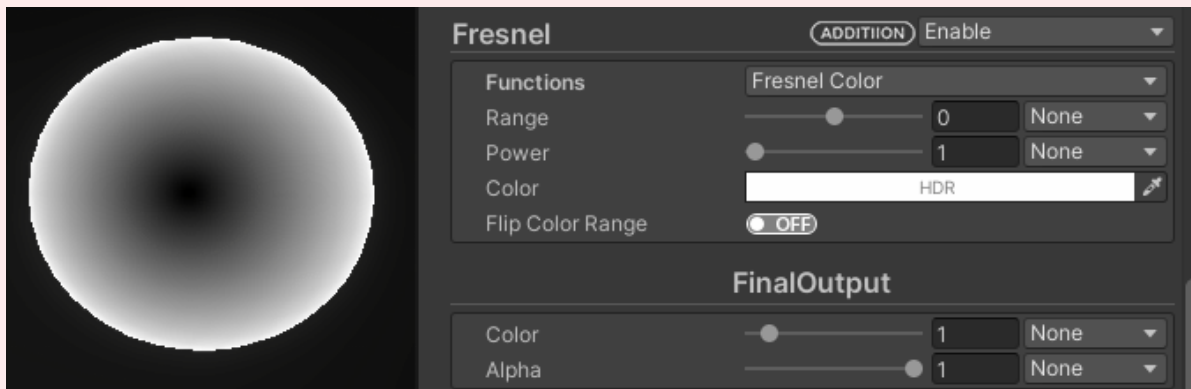
- **Range** - Controls the range of the fresnel.
- **Power** - Controls the intensity of the fresnel.
- **Color** - Controls the color of the fresnel.
- **Flip Color Range** - Reverses the range of the fresnel.

3 Fresnel Alpha



- Range - Controls the range of the fresnel.
- Power - Controls the intensity of the fresnel.
- Flip Alpha Range - Reverses the range of the fresnel.

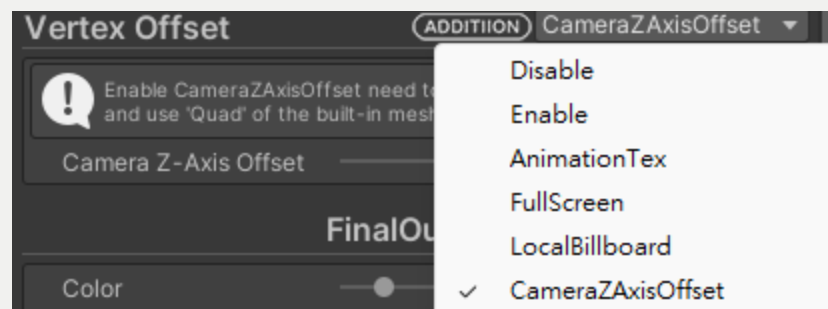
! Range and Power are shared parameters!



▼ Vertex Offset



Function Info



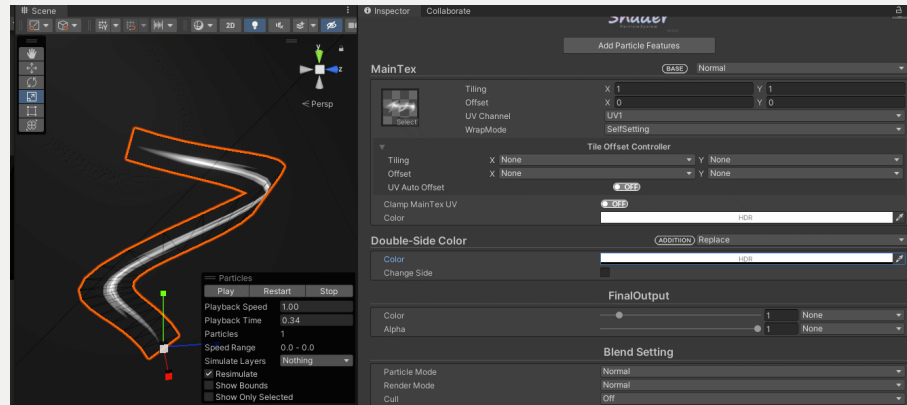
Function	Introduce	Example
Enable	Standard Vertex Mode	Vertex Offset (Enable).
AnimationTex	Vertex Animation Texture	Vertex Offset (Animation Tex).
FullScreen	Full Screen Vertex	Vertex Offset (FullScreen).
LocalBillboard	Local Billboard Vertex	Vertex Offset (Local Billboard).
CameraZAxisOffset	Camera Z-Axis Offset	Vertex Offset (Camera Z-axis Offset).

▼ Double-Side Color

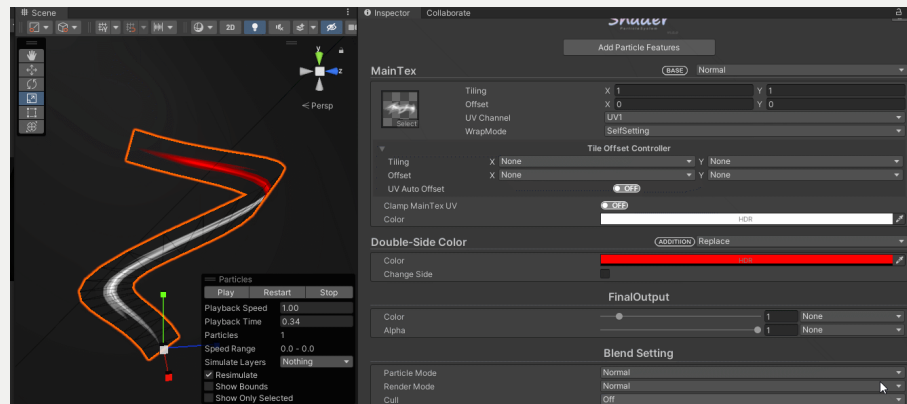


Function Info

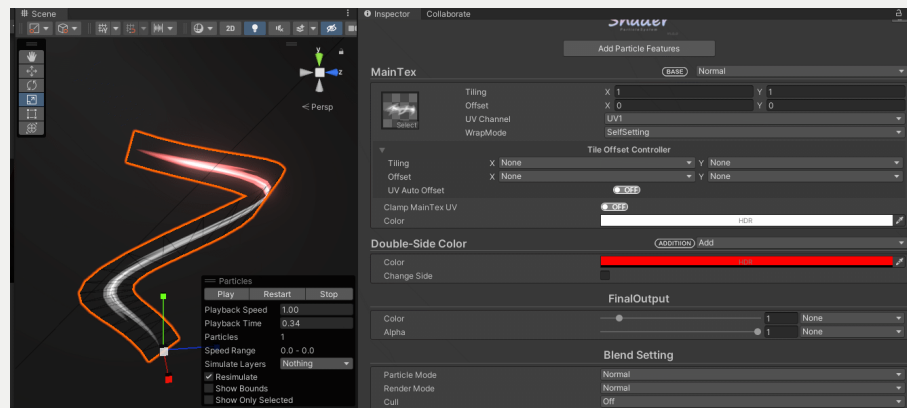
1 Replace - Complete Replacement



2 Additive - Brighten

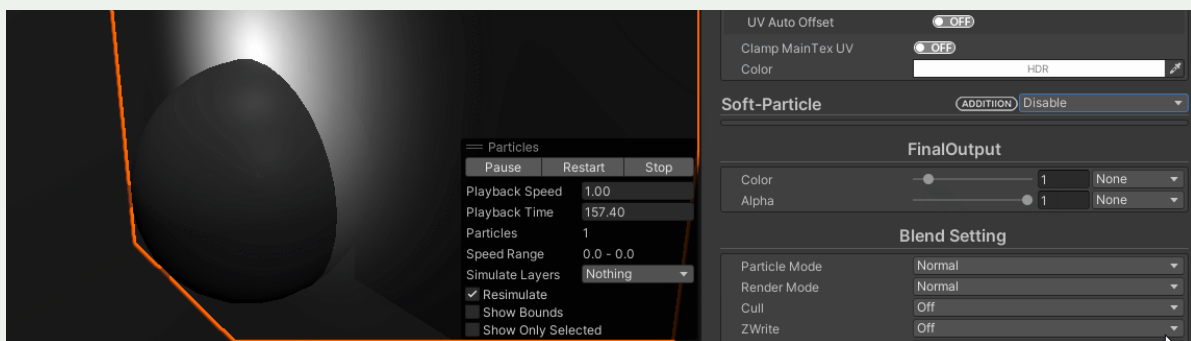
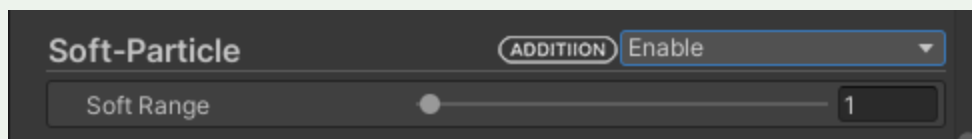


3 Multiply - Overlay



- Color - Controls color
- Change Side - Reverse Control Side

▼ Soft-Particle



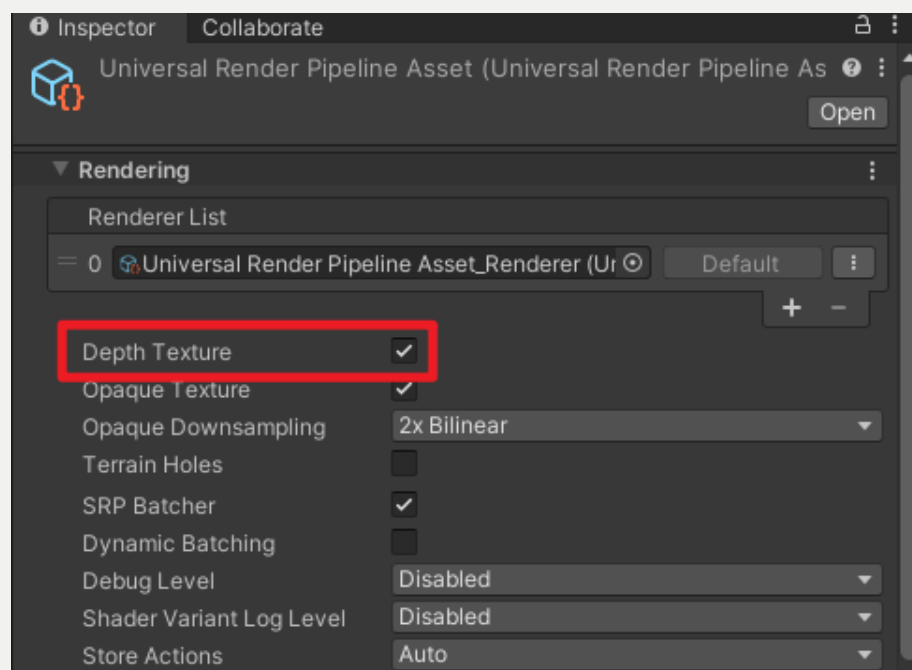
Example

Function Info

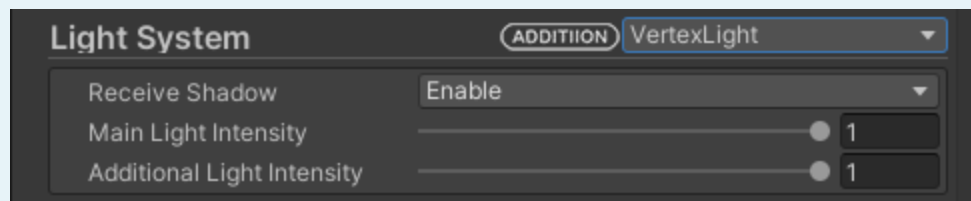
- Soft Range - Controls the intensity of the soft edges.

⚠ Precautions

- 1 • The scene must have '**Depth Texture**' mode enabled.

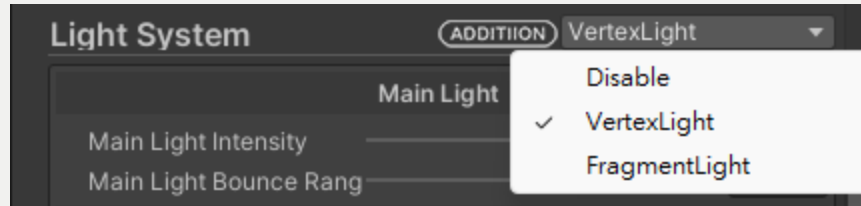


▼ Light System



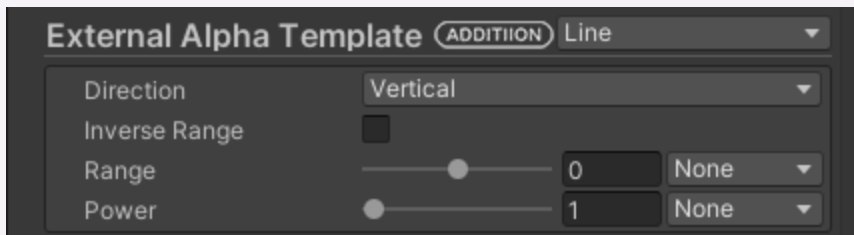
Function Info

▼ Light Mode

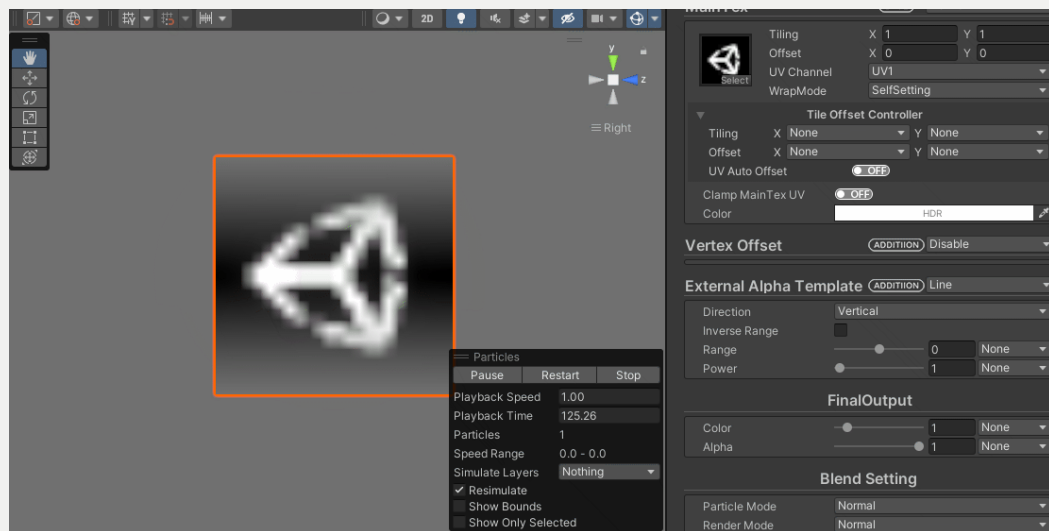
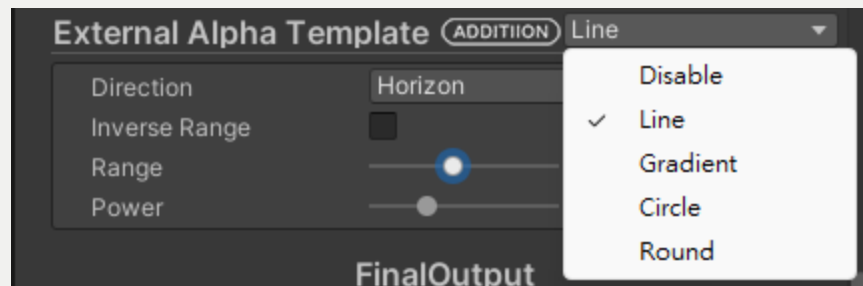


Function	Introduce	Example
Vertex Light	Vertex Lighting (Cheap)	Vertex Light
Fragment Light	Per-Pixel Lighting (Expensive)	Fragment Light

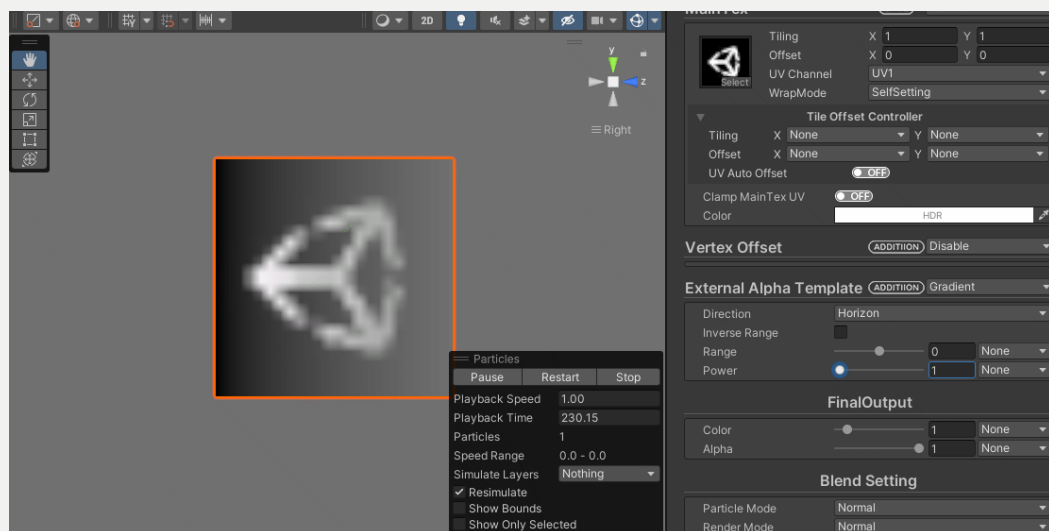
▼ External Alpha Template



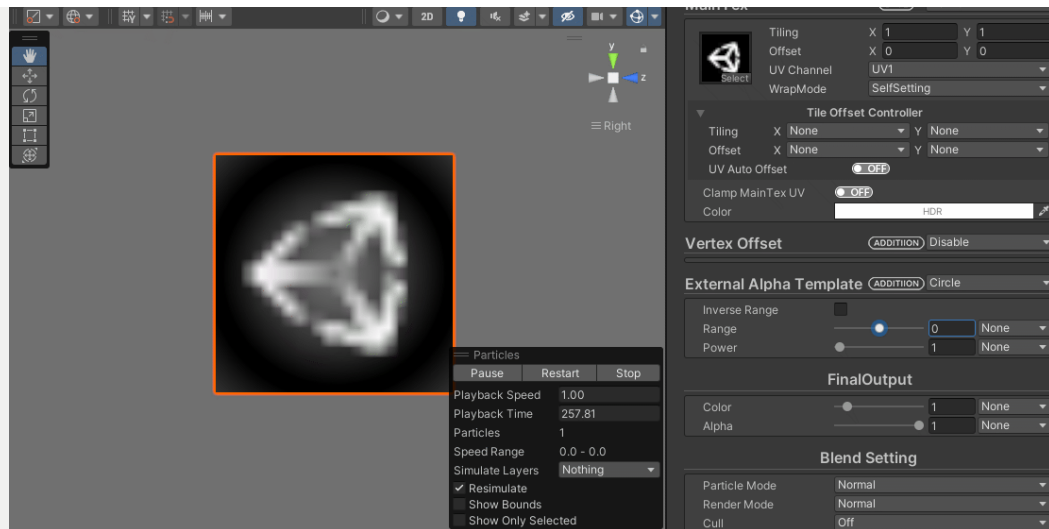
Function Info



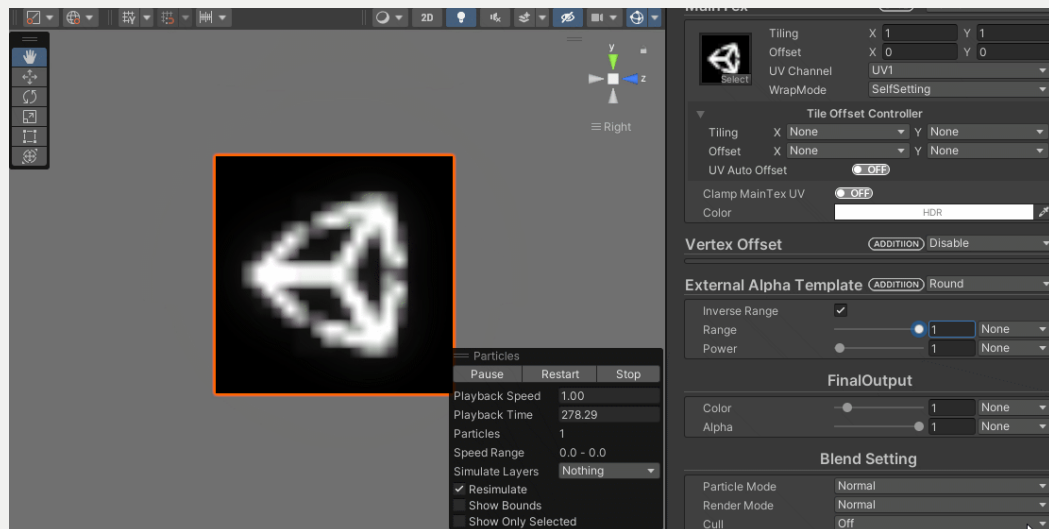
(Example) Template - Line



(Example) Template - Gradient



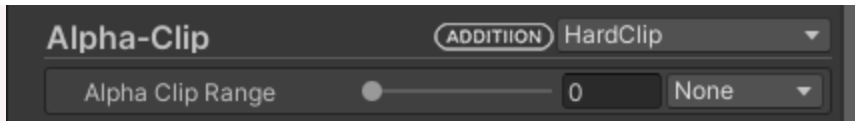
(Example) Template - Circle



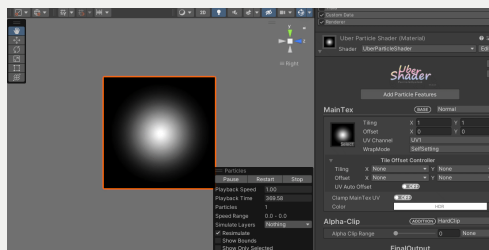
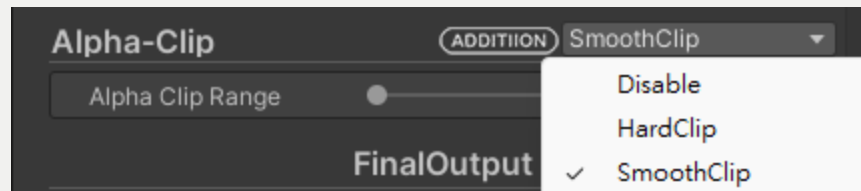
(Example) Template - Round

- **Direction** - Controls horizontal or vertical displacement.
- **Inverse Range** - Inverts the display range.
- **Range** - Controls the size of the range.
- **Power** - Controls the intensity of the range.

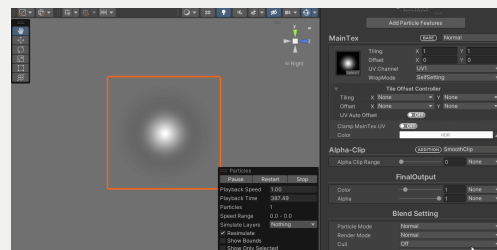
▼ Alpha-Clip



Function Info



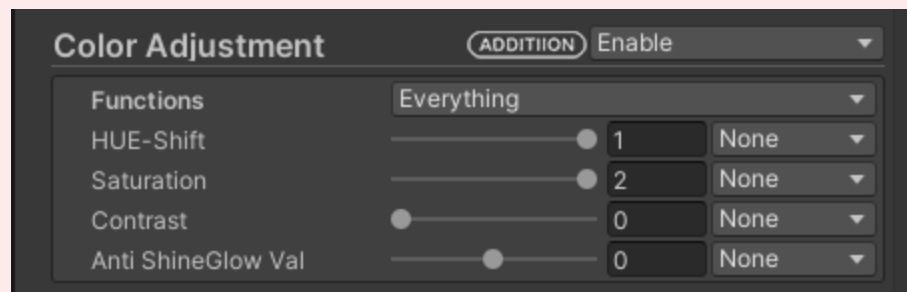
Example (Hard Clip)



Example (Smooth Clip)

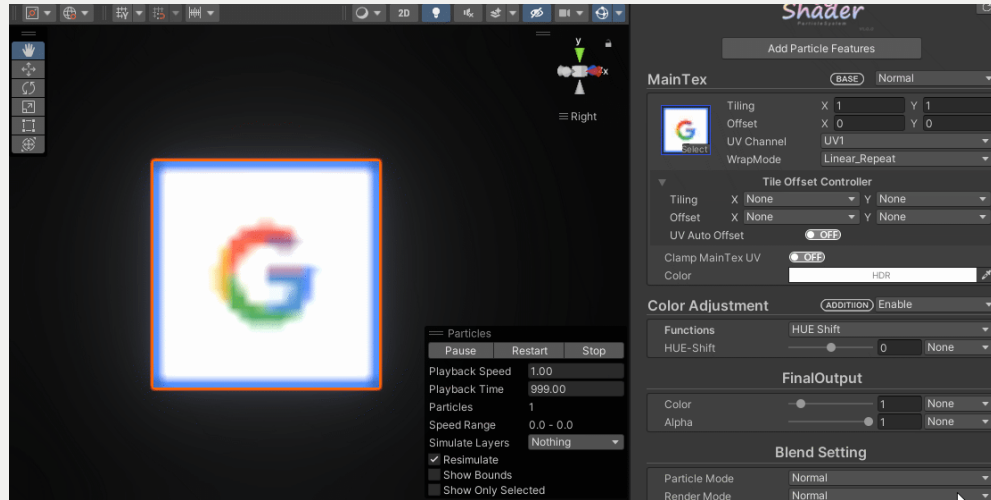
- Alpha Clip Range - Controls the transparency clipping range.

▼ Color Adjustment



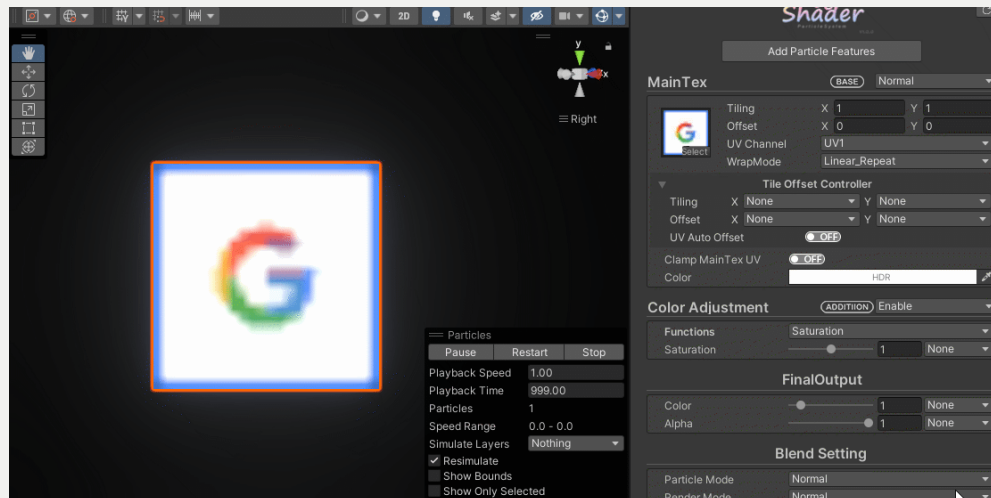
Function Info

▼ HUE-Shift



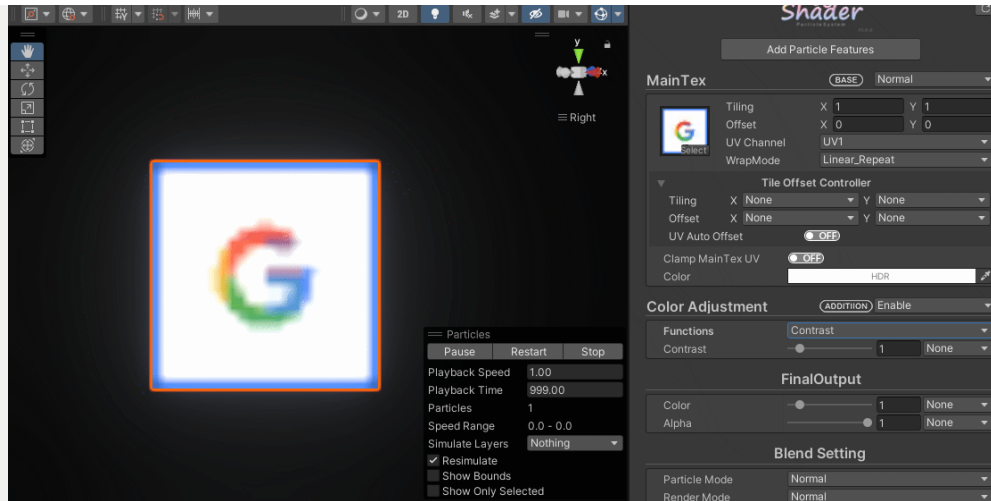
Example (HUE-Shift)

▼ Saturation



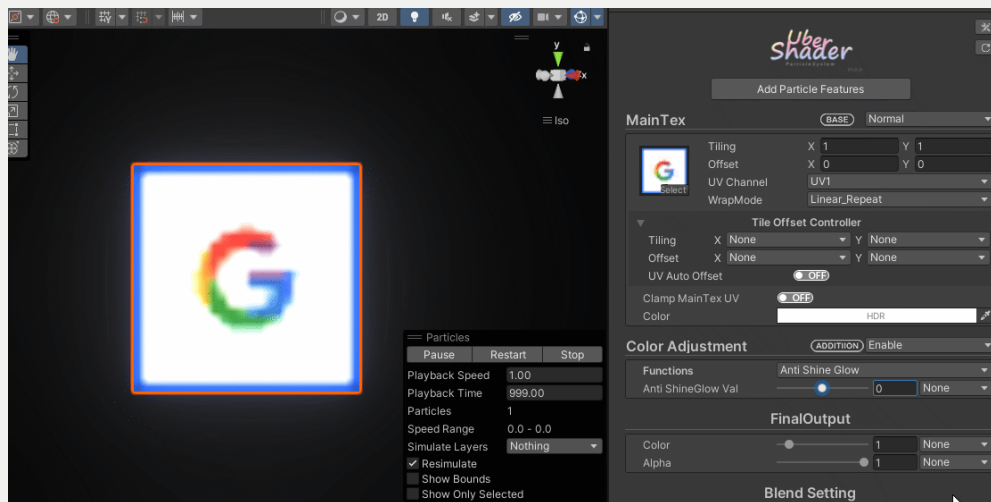
Example (Saturation)

▼ Contrast



Example (Contrast)

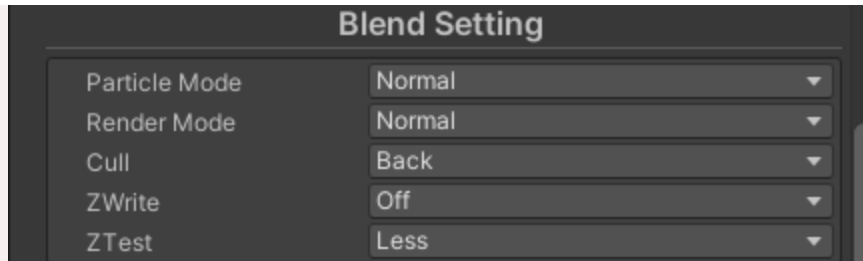
▼ Anti-ShineGlow



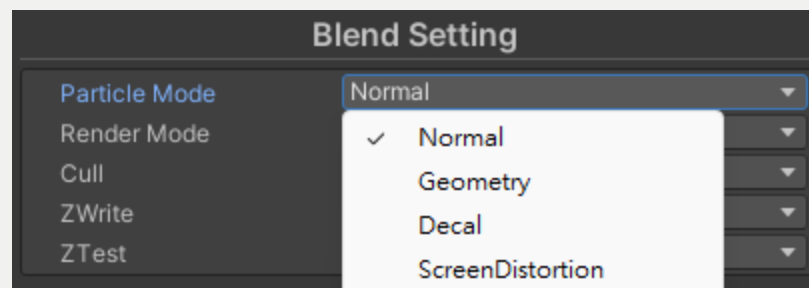
Example (Anti-ShineGlow)

Blend

▼ Blend Setting

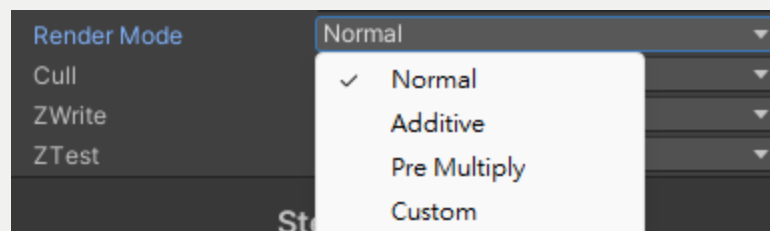


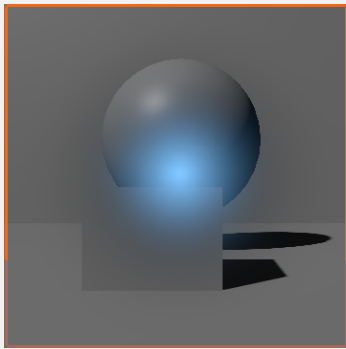
▼ Particle Mode



Function	Introduce	Example
Normal	Standard Particles (default translucent)	None
Geometry	Opaque Particles	Particle Mode - Geometry
Decal	Decal Particles	Particle Mode - Decal
Screen Distortion	Screen Distortion Particles	Particle Mode - Screen Distortion

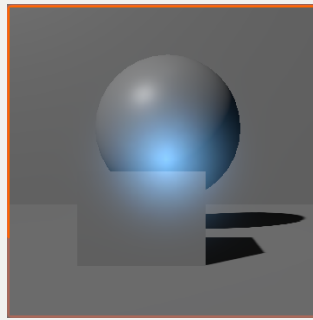
▼ Render Mode





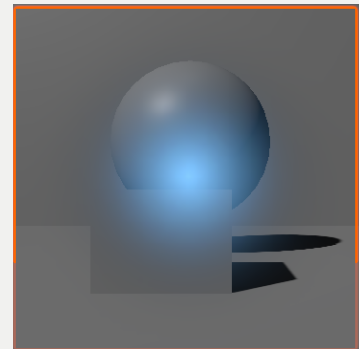
Normal

$(SrcAlpha + OneMinusSrcAlpha)$



Additive

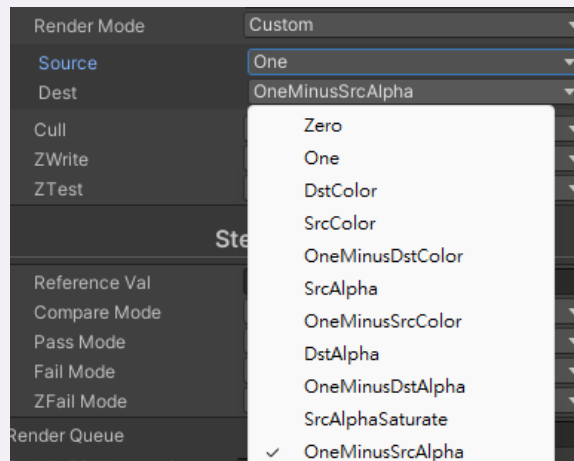
$(SrcAlpha + One)$



Pre Multiply

$(One + OneMinusSrcAlpha)$

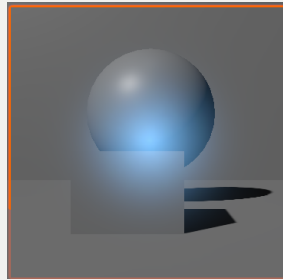
▼ Custom



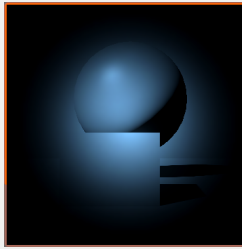


By selecting different Source and Destination settings, various types of layer attributes can be achieved.

Common Use



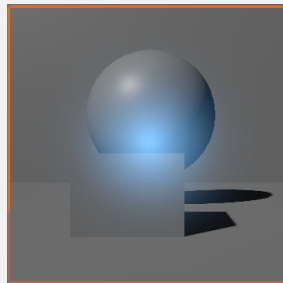
Soft Additive |
 $\text{OneMinusDstColor} + \text{One}$



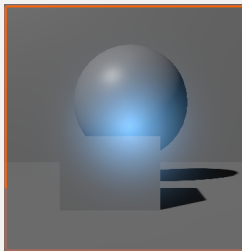
Multiply | $\text{DstColor} + \text{Zero}$



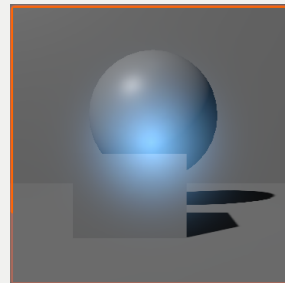
2x Multiply | $\text{DstColor} + \text{SrcColor}$



Screen |
 $\text{OneMinusDstColor} + \text{One}$

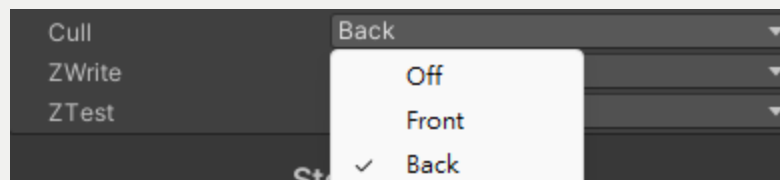


Screen | $\text{One} + \text{OneMinusSrcColor}$

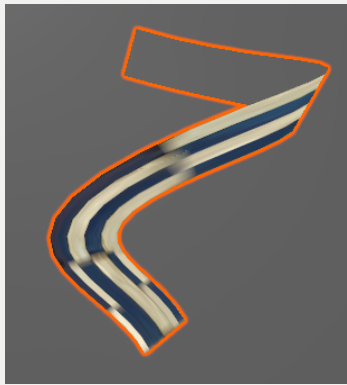


Linear Dodge | $\text{One} + \text{One}$

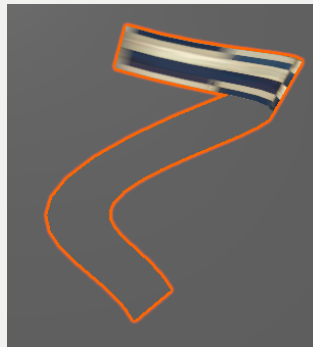
▼ Cull



Cull Module



Back

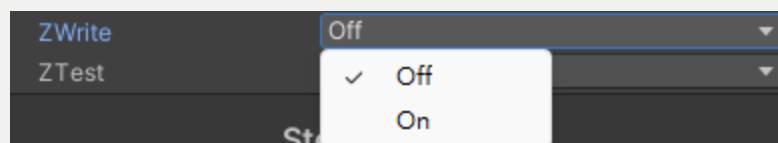


Front

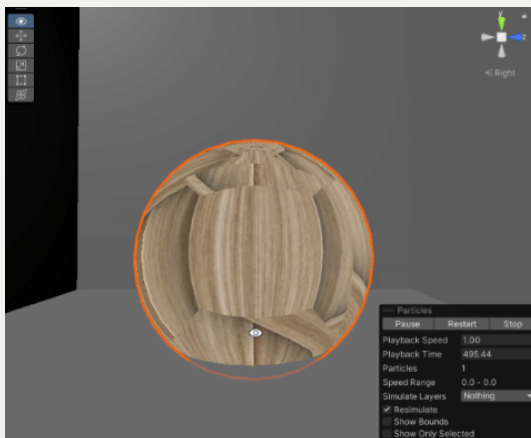


Off

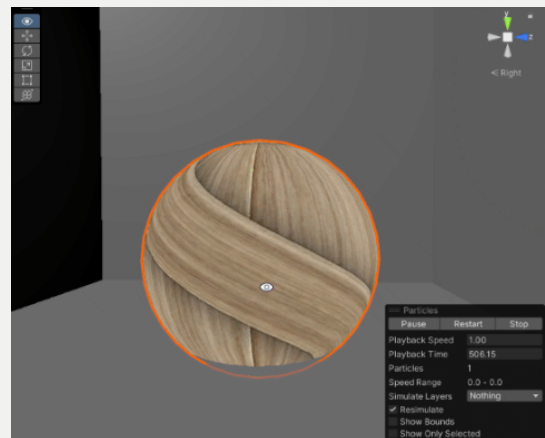
▼ ZWrite



ZWrite Module

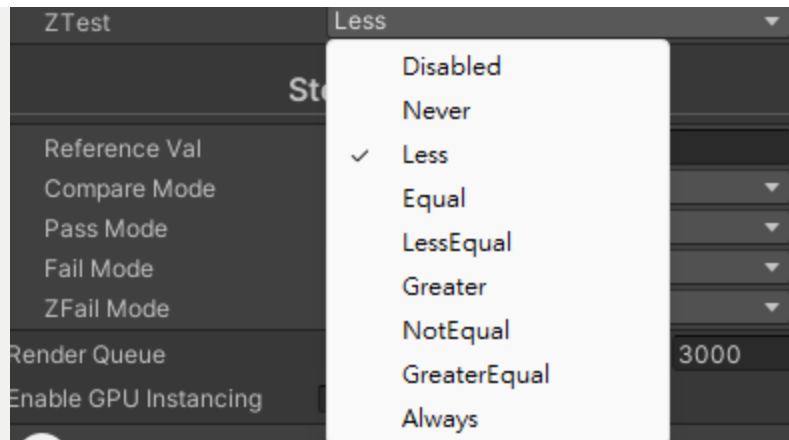


ZWrite Off

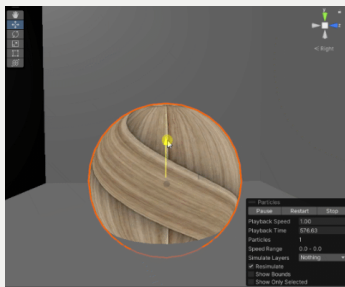


ZWrite On

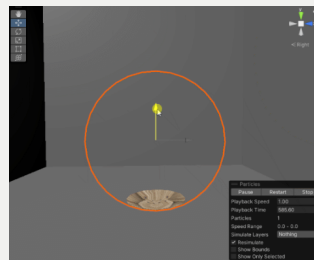
▼ ZTest



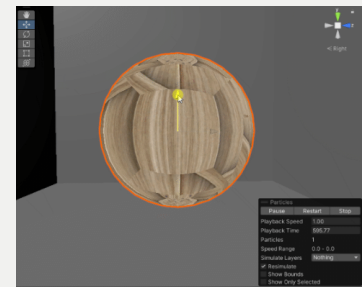
ZTest Module



Less / LessEqual



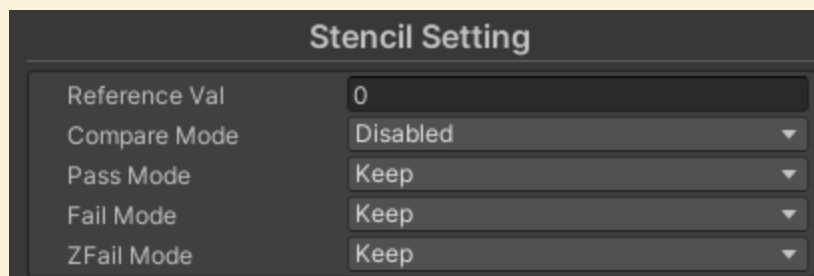
Greater / GreaterEqual



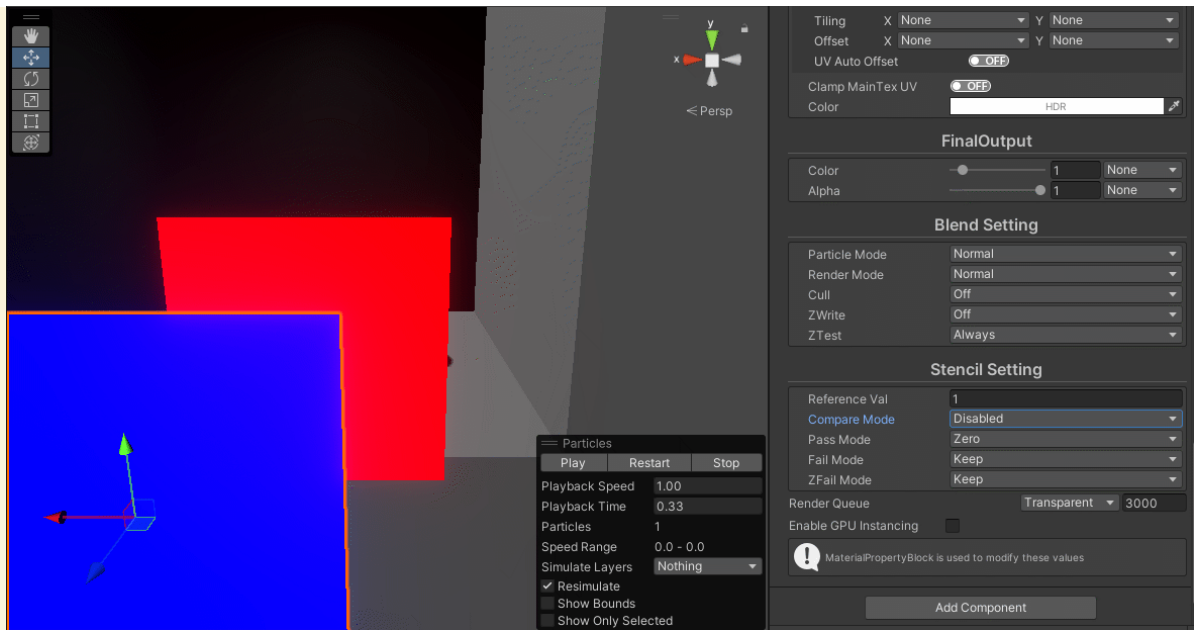
NotEqual / Always

Stencil

▼ Stencil Setting



Stencil Module

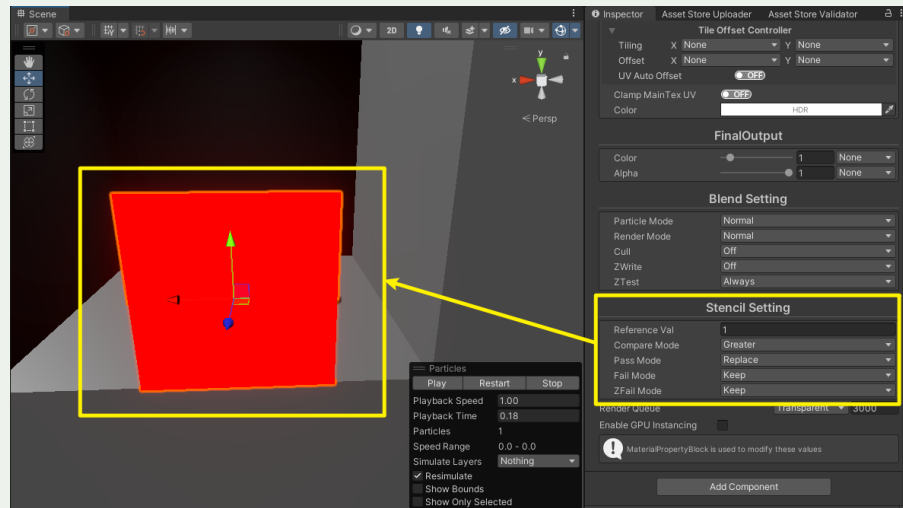


Example

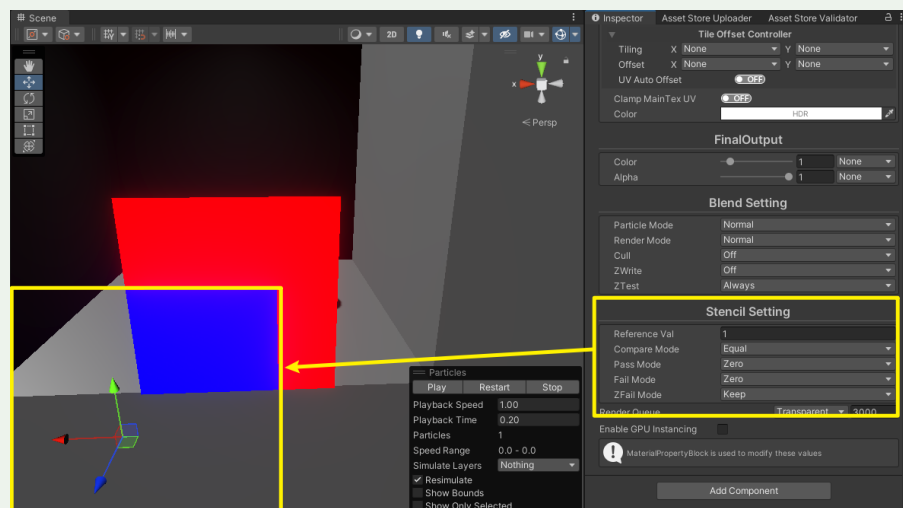
? What is a Stencil Buffer?



Simply put, a stencil buffer allows pixels that successfully write into the color buffer to be assigned a value, providing a reference ID for subsequent objects that want to use masks. Under normal circumstances, the stencil ID for all pixels in the color buffer is set to 0.



A red stencil uses the value 1 to compare with the stencil ID in the color buffer; if the comparison passes, it will replace the value in the buffer.



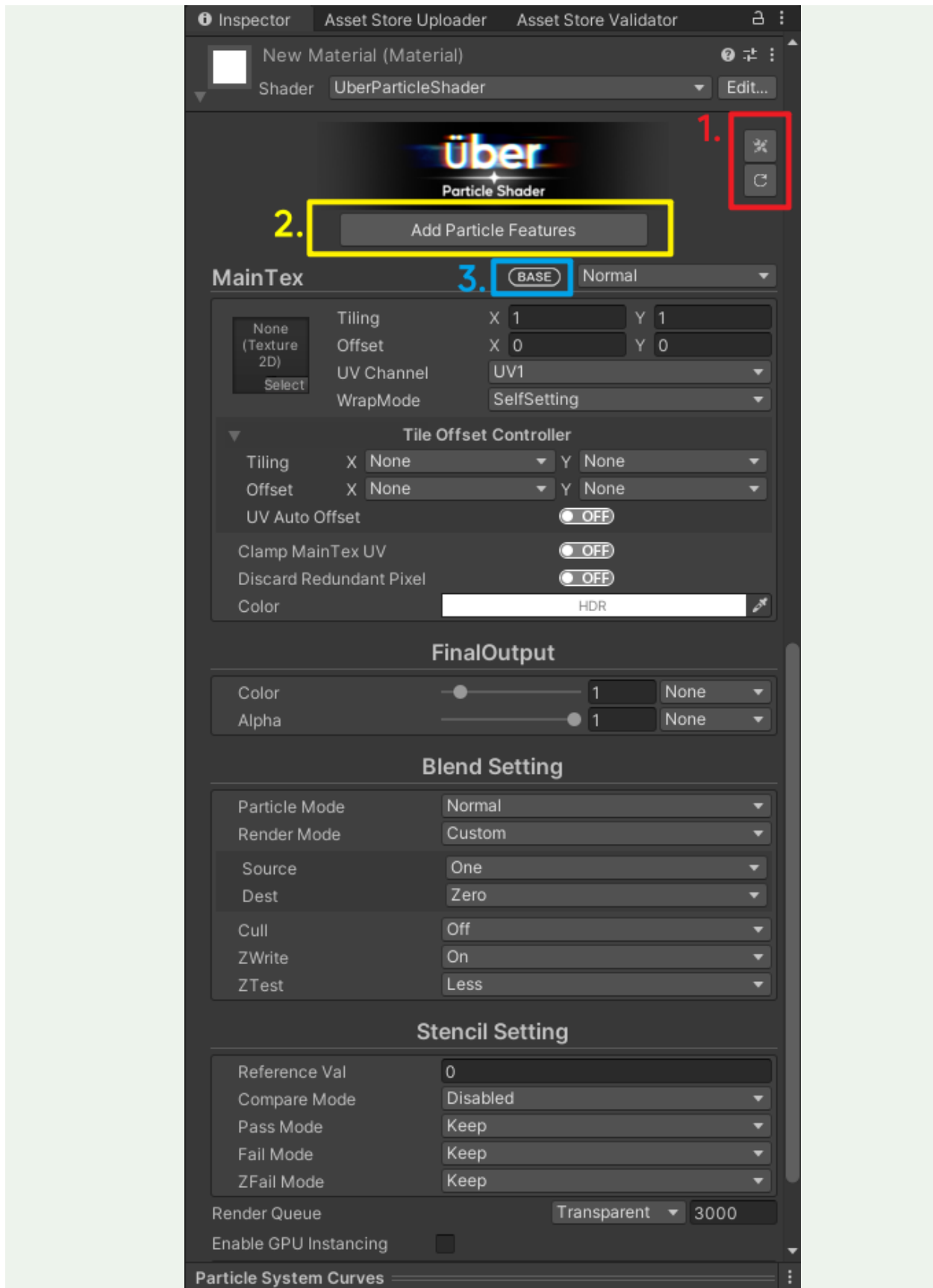
A blue stencil also uses the value 1 for comparison with the stencil ID in the color buffer; if the comparison passes, it will reset the buffer's value to zero.

? Why reset the stencil to zero?

Using a stencil to create masks can cause issues with ID matching for subsequent users who are creating special effects with stencils if it is not cleared back to zero.

Other

▼ Inspector Setting



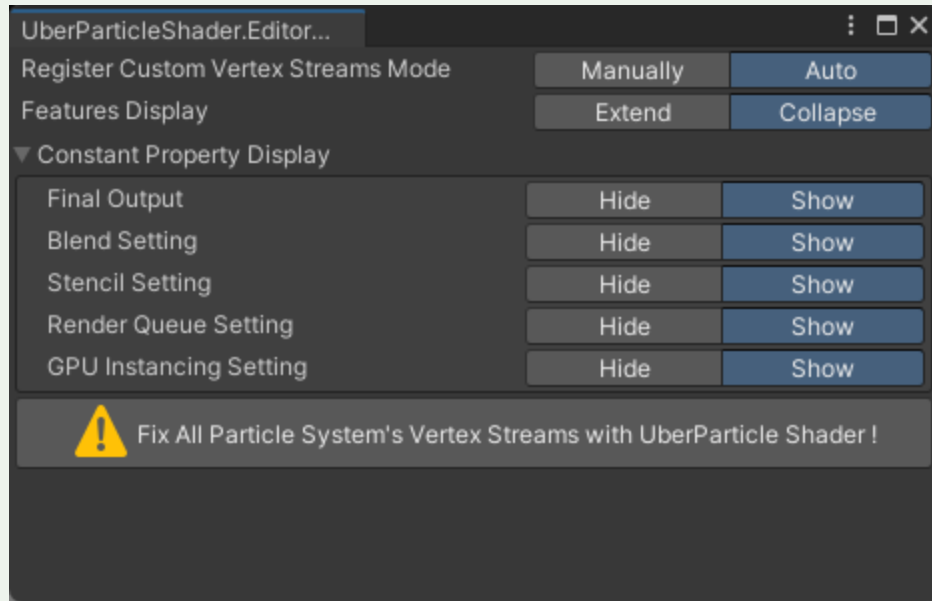
UberParticleSystem Default Inspector

▼ Side Buttons

Personalized Settings



Clicking the button will automatically launch this interface, where you can change the interface settings for UberShaderParticle through the settings below.

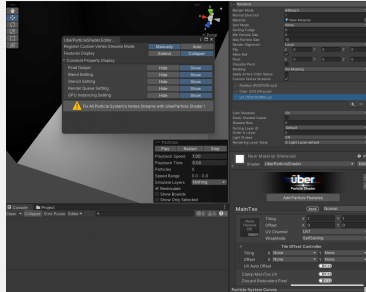


Personalization Settings Panel

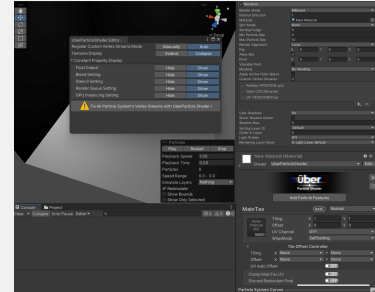
▼ Register Custom Vertex Streams Mode - **Manual**/**Automatic**
Custom Vertex Data



You can choose to manually or automatically control custom vertex data. (Automatic is recommended)



Manual

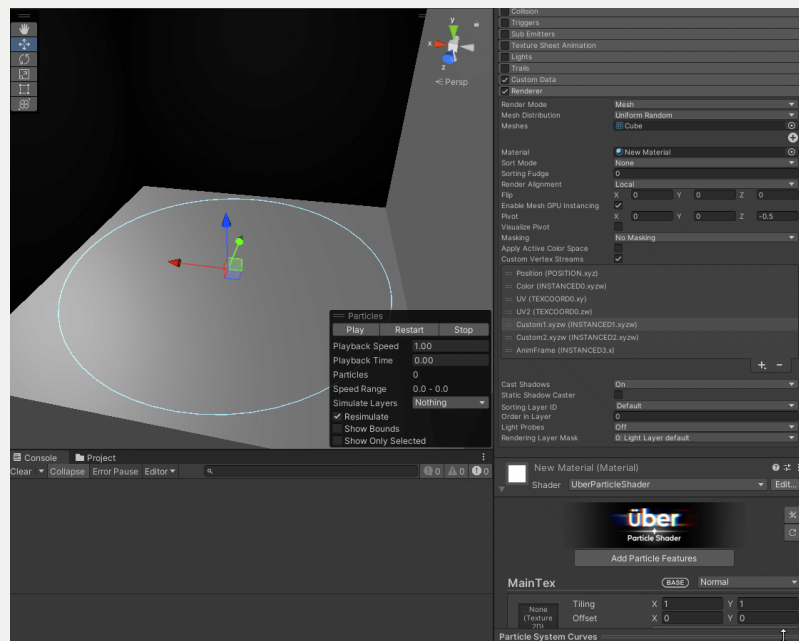


Automatic

? Why control custom vertex data?

Before enabling 'Enable Mesh GPU Instance', custom vertex data is solely responsible for passing parameters. Even if the parameters are mistakenly assigned, at most, the effects will not appear, and it won't cause any significant issues.

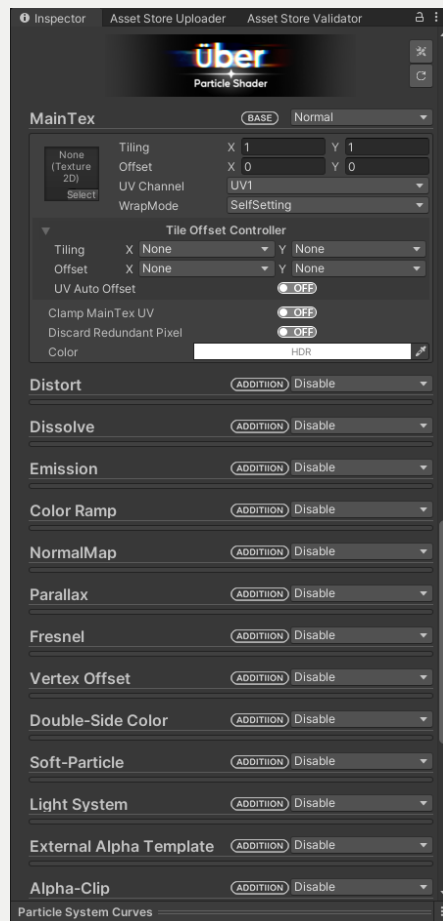
However, once '**Enable Mesh GPU Instance**' is used, all vertex data of the model will be handled by the GPU. To speed up vertex production, **this feature implements a unified format for vertex data**, saving the process of individually specifying model vertex data. **If the vertex data of a particle effect is incorrect**, it may cause flickering, disappearance, or other anomalies.



Abnormal displays that occur when 'Enable Mesh GPU Instance' is enabled without properly providing vertex data

▼ Feature Display - Mode of displaying integrated functions

Extend



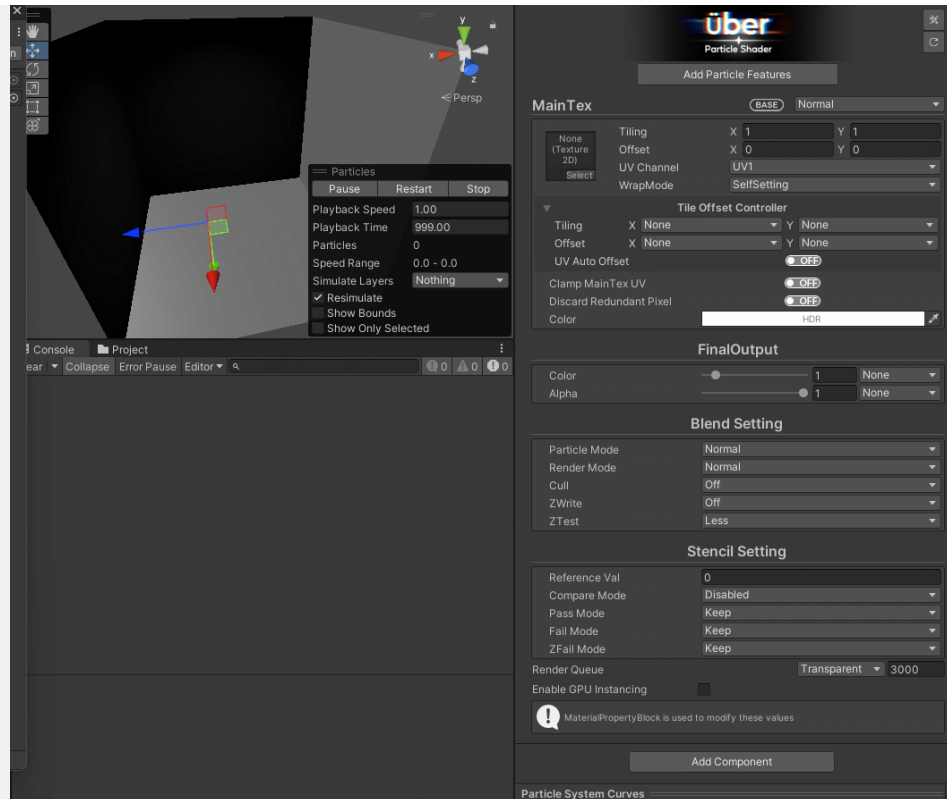
Expand all features

Collapse



Collapse all features and add new ones through a 'menu-style'

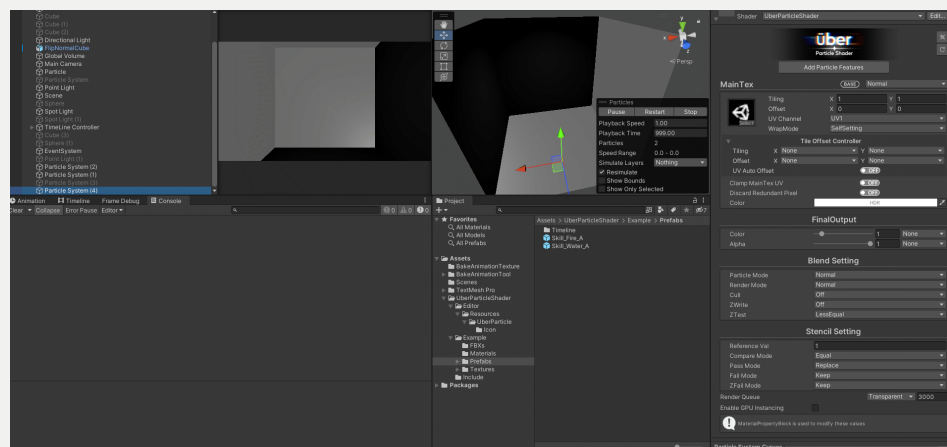
▼ Constant Property Display - Fixed parameter panel display mode



Merge / Store parameter panel

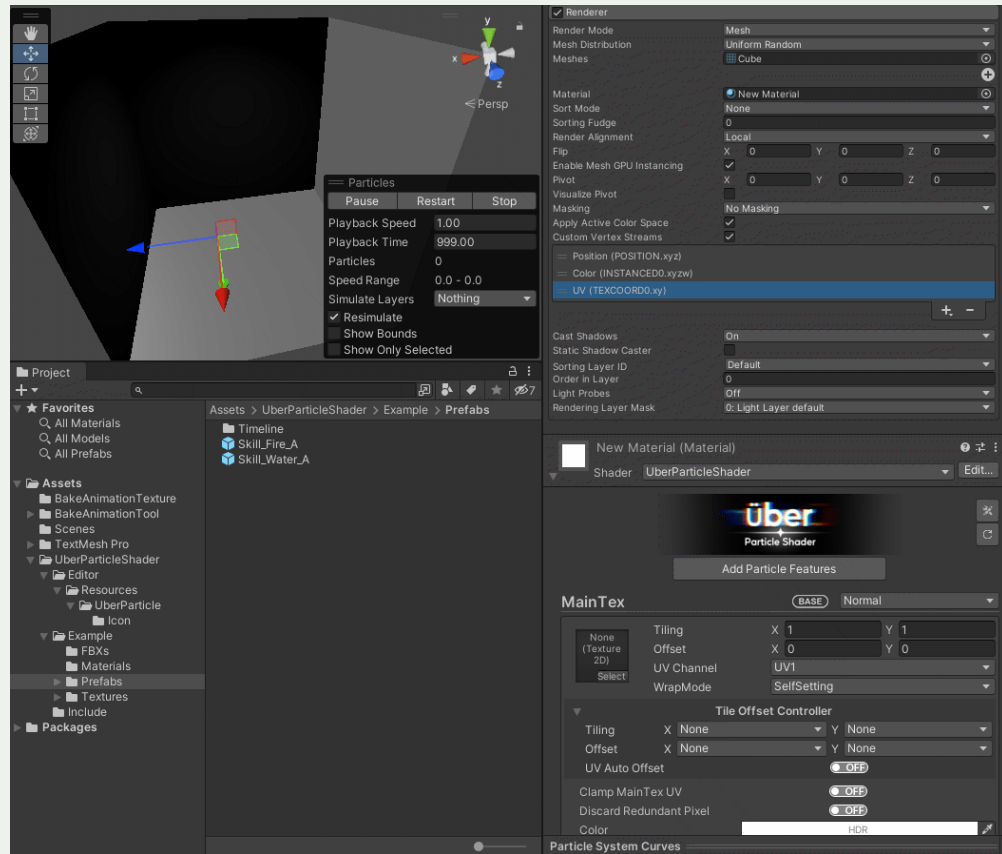
▼ Fix All Particle System's Vertex Steams with UberParticle Shader !

- Refresh all custom vertex data for objects using the UberParticle Shader



Refresh all effects using the UberParticle Shader (only for 'vertex data' settings)

Calibration button (Custom Vertex Stream)



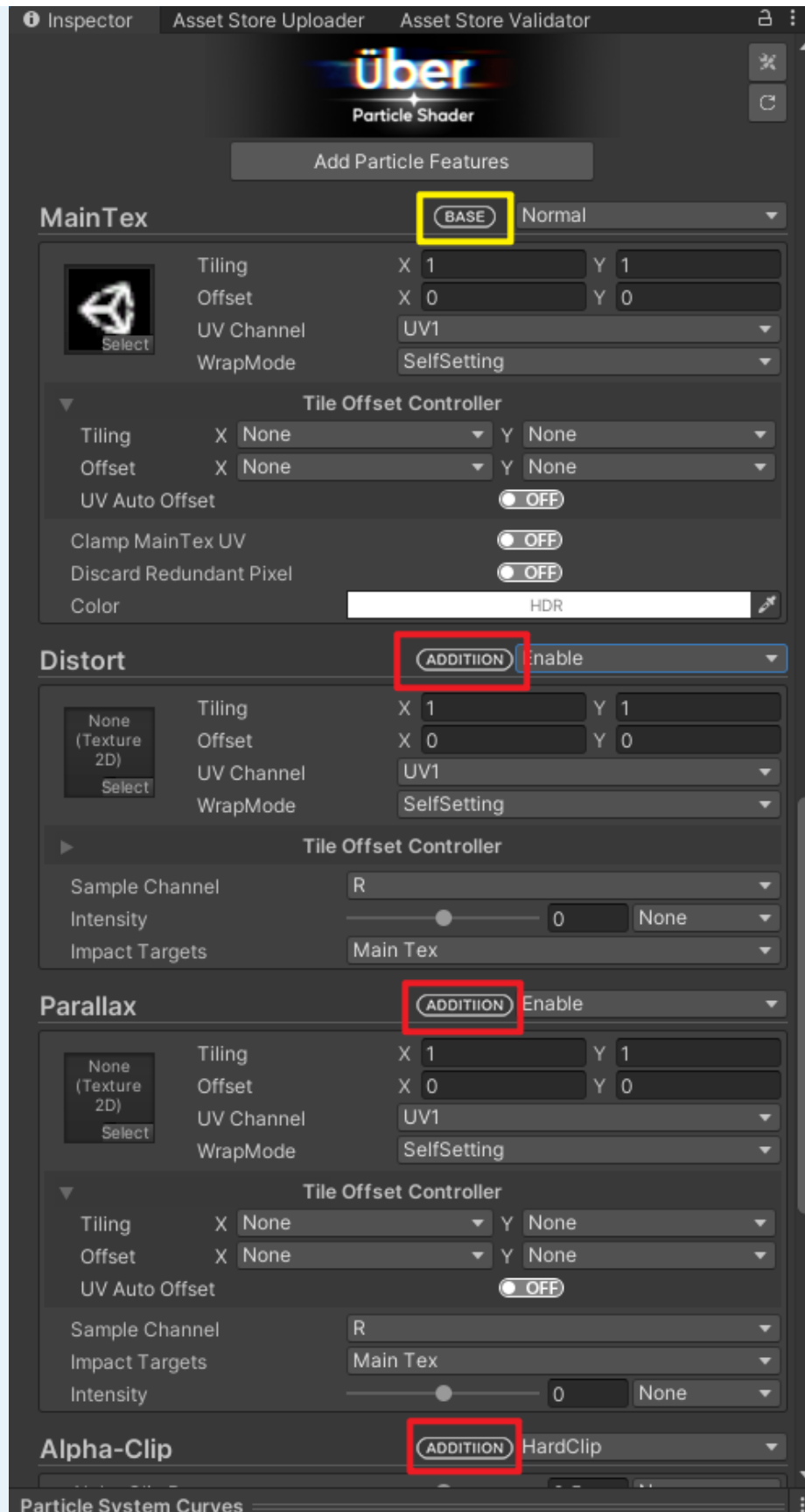
Reset 'current effect' vertex data

▼ Add Particle Features



Add menu function

▼ Tags



Feature Tabs

- Base - Basic features (cannot be removed)
- Addition - Additional features

▼ Q&A

▼ Reference Pages

[Multi-Channel Tex](#)

[FlipBook](#)

[Polar Coord](#)

[ScreenUV](#)

[Flow Map](#)

[Parallax \(Enable\)](#)

[Parallax \(Steep\)](#)

[Vertex Offset \(Enable\)](#)

[Vertex Offset \(Animation Tex\)](#)

[Vertex Offset \(FullScreen\)](#)

[Vertex Offset \(Local Billboard\)](#)

[【Tool】 BakeAnimationTool](#)

[Vertex Offset \(Camera Z-axis Offset\)](#)

[Vertex Light](#)

[Fragment Light](#)

[Particle Mode - Geometry](#)

[Particle Mode - Decal](#)

Particle Mode - Screen Distortion